Windows Event Logging and Monitoring

Topic: Event Logging and Monitoring Lab
Task: Generate Windows Event Log and Monitor them

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1. Objective

The objective of this practical is to understand Windows Security Event Logging and learn how to generate, locate, and analyse important Windows events.

In this practical, I will generate and investigate the following events:

1. Failed Logon - Event ID 4625
2. Successful Logon - Event ID 4624
3. New User Creation - Event ID 4720
4. User Added to Administrators Group - Event ID 4732
5. Service Creation - Event ID 7045
6. Security Log Cleared - Event ID 1102

The purpose is to understand how Windows records security-related activities and how a SOC analyst can use these events to investigate suspicious behaviour.

2. Lab Environment

Component	Details
Target Machine	Windows VM
Operating System	Windows 10/11
Main Tool	Windows Event Viewer
Secondary Tool	PowerShell
Command-Line Tool	Command Prompt
Log Location	Windows Event Viewer
Main Log	Security
Additional Log	System

Note: All activities are performed inside my own isolated Windows virtual machine.

3. Introduction to Windows Event Logging

Windows Event Logging is a built-in mechanism that records activities occurring within the operating system.

These records help administrators and security analysts monitor system activities, investigate problems, and detect suspicious behaviour.

Windows maintains three major event logs:

Log	Purpose	Examples
Application	Records application-related events	Application errors and crashes
Security	Records security and auditing activities	Logons, account creation, privilege changes
System	Records operating system and service events	Service installation, driver errors, system startup

For this practical, the Security and System logs are particularly important.

4. Opening Windows Event Viewer

Step 1: Launch Event Viewer

a. Press Windows + R.
b. Type eventvwr.msc.
c. Press Enter.
d. Event Viewer will open.

Step 2: Navigate to Security Logs

Follow this path:
Event Viewer → Windows Logs → Security

Step 3: Filter Current Log

a. Right-click Security.
b. Select Filter Current Log.
c. Enter the required Event ID in the field named All Event IDs.
d. Click OK.
e. Review the matching event records.

For multiple event IDs, use commas.
Example: 4624,4625,4720,4732

For Event ID 7045, navigate to:
Event Viewer → Windows Logs → System

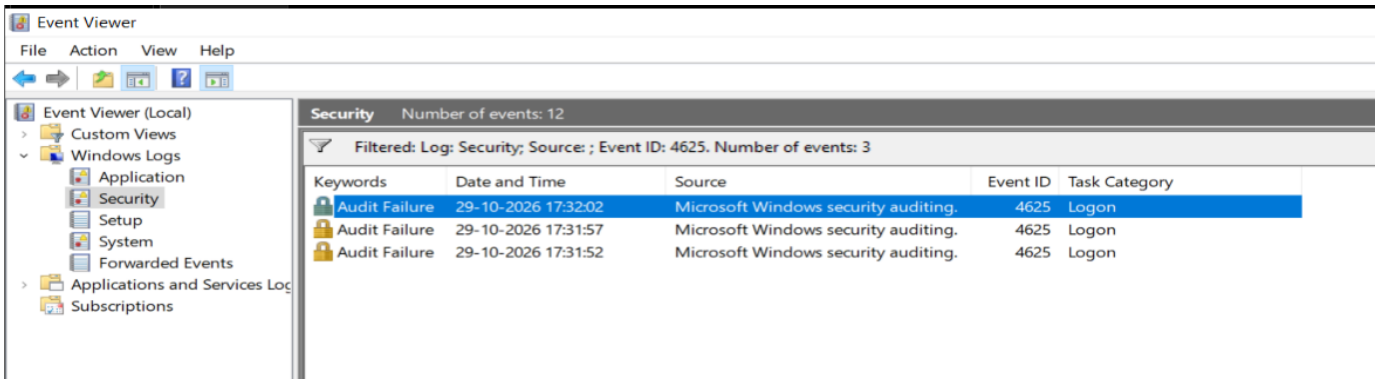
5. Practical Activities - Generating Windows Events

5.1.Generate Failed Logon - Event ID 4625

Objective: To generate a failed login attempt and identify how Windows records unsuccessful authentication.

Procedure:

a. Press Windows + L to lock the Windows VM.
b. At the login screen, enter an incorrect password for an existing local account.
c. Submit the incorrect password.
d. Return to Windows using the correct password if necessary.



Event Viewer Steps:

a. Open Event Viewer.
b. Navigate to Windows Logs → Security.
c. Click Filter Current Log.
d. Enter Event ID: 4625
e. Click OK.

Security Importance:

Repeated failed logon attempts may indicate:

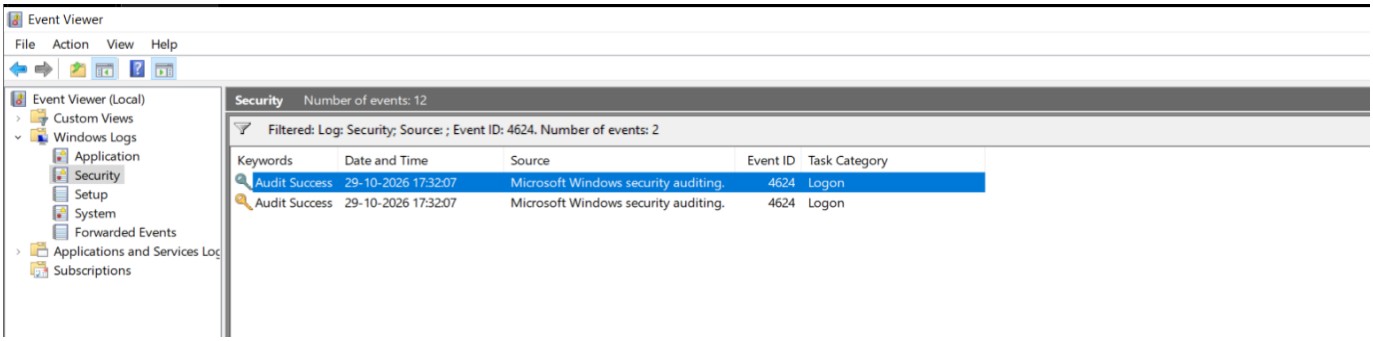
- Brute-force attacks
- Password guessing
- Unauthorised login attempts
- Suspicious authentication activity

5.2.Generate Successful Logon - Event ID 4624

Objective: To understand how Windows records successful authentication.

Procedure:

a. Lock the Windows VM using Windows + L.
b. Unlock it using the correct username and password.
c. Alternatively, sign out and sign back in to generate a fresh interactive logon event.



Event Viewer Steps:

a. Open Event Viewer.
b. Navigate to Windows Logs → Security.
c. Select Filter Current Log.
d. Enter: 4624
e. Click OK.
f. Open the relevant event.

Security Importance:

This event helps analysts:

- Track successful user authentication.
- Identify unexpected login times.
- Investigate unusual login sources.
- Correlate successful logins with previous failed attempts.

5.3.Generate New User - Event ID 4720

Objective:

To create a new local Windows user and observe the corresponding security event.

Procedure:

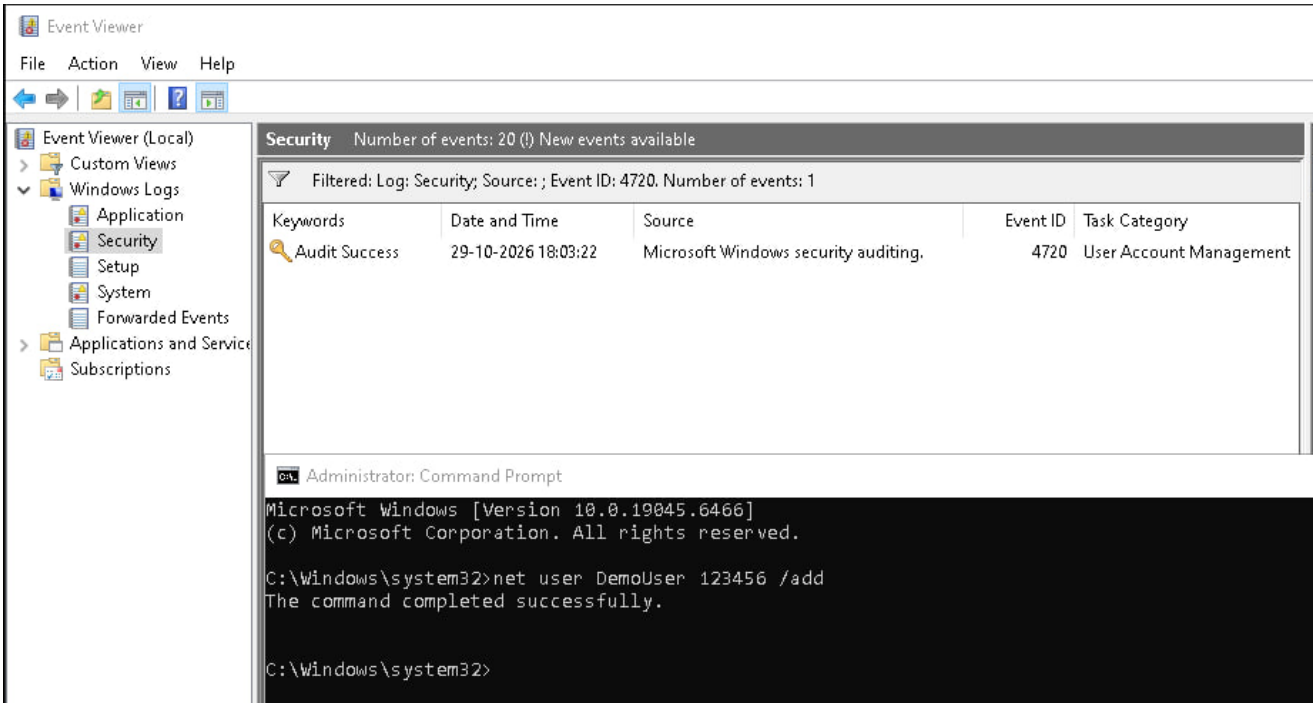
1. Open Command Prompt or PowerShell as Administrator.
2. Execute:

net user DGTestUser P@ssw0rd123! /add

3. Windows will create a local user account named DGTestUser.

Event Viewer Steps:

a. Open Event Viewer.
b. Navigate to Windows Logs → Security.
c. Click Filter Current Log.
d. Enter: 4720
e. Click OK.
f. Open the event generated during account creation.



Security Importance:

Unexpected account creation may indicate:

- Persistence attempts by attackers.
- Creation of unauthorised accounts.
- Suspicious administrative activity.

5.4.Generate User Added to Administrators - Event ID 4732

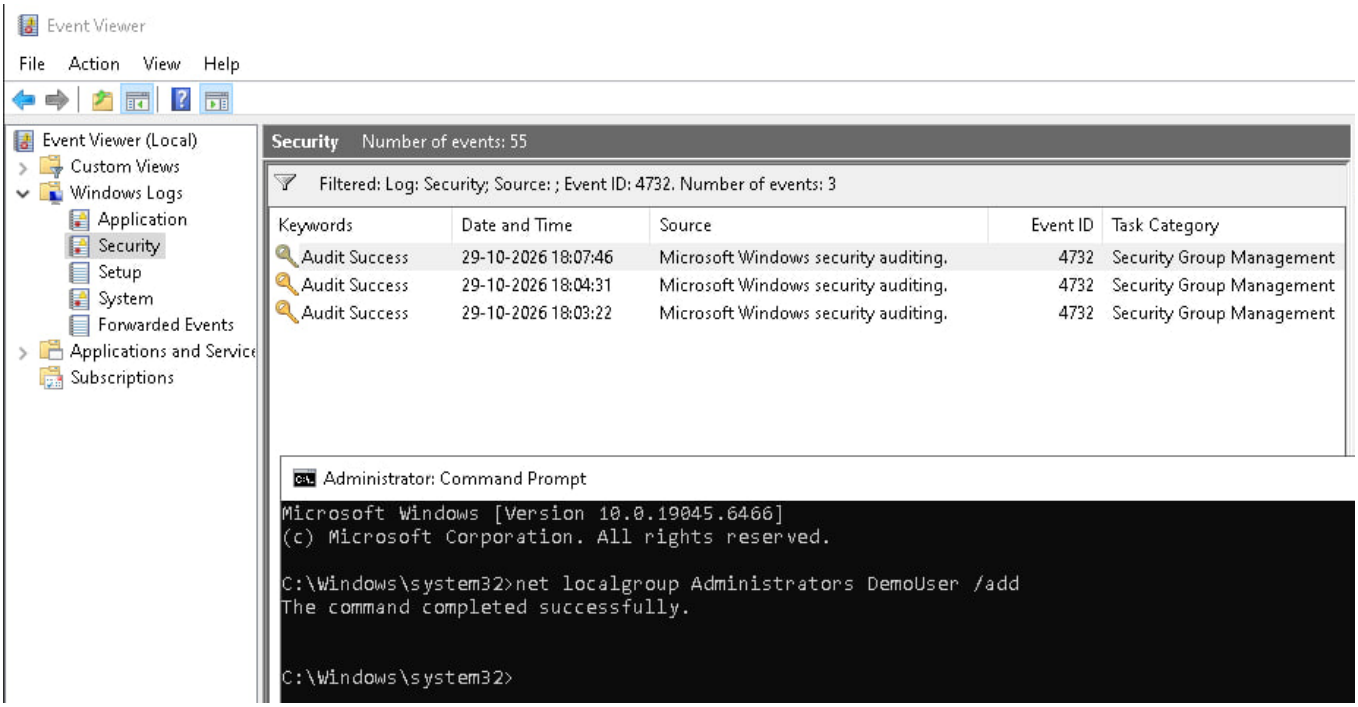
Objective: To add the newly created user to the local Administrators group and observe the corresponding security event.

Procedure:

a. Open Command Prompt as Administrator.
b. Execute: net localgroup Administrators DGTestUser /add
c. Windows will add DGTestUser to the local Administrators group.

Event Viewer Steps:

a. Navigate to Windows Logs → Security.
b. Click Filter Current Log.
c. Enter: 4732
d. Click OK.
e. Open the relevant event.



Security Importance:

Adding an account to the Administrators group gives it elevated privileges.
Unexpected changes may indicate:

- Privilege escalation.
- Unauthorised administrative access.
- Attacker persistence.
- Misuse of administrator privileges.

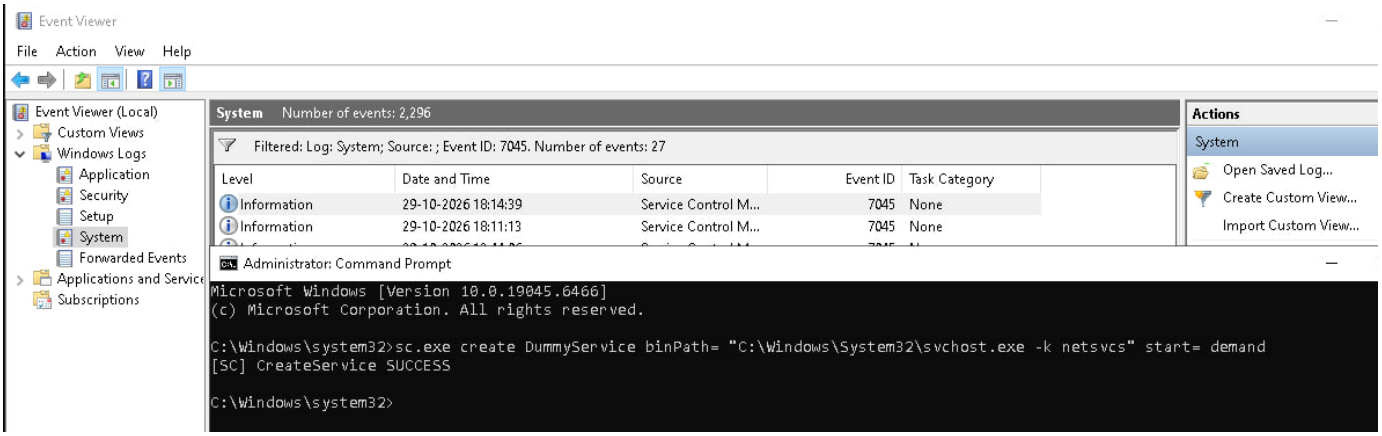
5.5.Generate Service Installation - Event ID 7045

Objective: To create a test Windows service and observe the service installation event.
Procedure:

a. Open Command Prompt as Administrator.
b. Execute: sc.exe create DGTestService binPath= "C:\Windows\System32\svchost.exe -k netsvcs" start= demand
c. The command creates a test service named DGTestService.
d. Starting the service is not necessary for generating the installation event.

Event Viewer Steps:

a. Open Event Viewer.
b. Navigate to: Windows Logs → System
c. Click Filter Current Log.
d. Enter: 7045
e. Click OK.
f.
g. Locate the event associated with DGTestService.



Security Importance:

Attackers may install malicious services to:

- Maintain persistence.
- Execute malicious programs.
- Establish unauthorised background processes.
- Maintain access after system restarts.

After collecting the evidence, remove the test service:

sc.exe delete DGTestService

This removes the test service registration. The original 7045 installation event should remain in the System log.

5.6.Generate Security Log Cleared - Event ID 1102

Objective: To understand how Windows records the clearing of the Security event log.

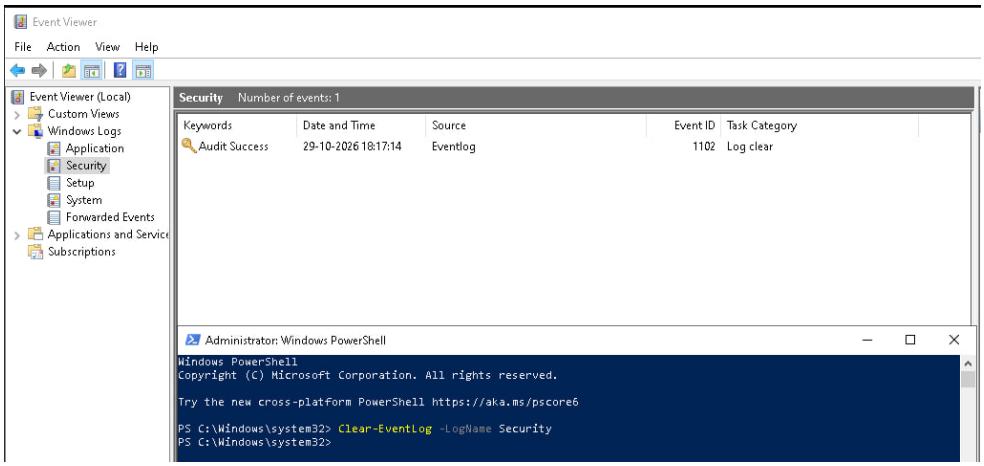
Important Safety Note: Clearing the Security log removes existing records and can destroy investigation evidence. Perform this activity only in your isolated lab VM. Export or preserve any required logs before proceeding.

Procedure:

a. Open Event Viewer as Administrator.
b. Navigate to Windows Logs → Security.
c. Right-click Security.
d. Select Clear Log.
e. If you need to preserve the existing logs, choose Save and Clear.
f. Confirm the operation.

Event Viewer Steps:

a. Reopen Windows Logs → Security.
b. Select Filter Current Log.
c. Enter: 1102
d. Click OK.
e. Locate the event generated by clearing the Security log.



Security Importance:

Clearing the Security log may indicate:

- An attempt to remove evidence.
- Anti-forensic activity.
- An attacker attempting to hide their actions.
- Unauthorised administrative activity.

6. Summary of Generated Events

Event ID	Activity	Windows Log
4625	Failed logon	Security
4624	Successful logon	Security
4720	New user created	Security
4732	User added to Administrators	Security

7045	New service installed	System
1102	Security log cleared	Security

7. SOC Analyst Perspective

Windows Security Event Logs are an important source of information for Security Operations Centre (SOC) analysts.

By monitoring these events, analysts can identify suspicious activities and investigate possible security incidents.

Some important detection scenarios include:

Scenario 1: Brute-Force Detection

- Monitor repeated Event ID 4625 records.
- Identify the account being targeted.
- Check the source IP address.
- Correlate failed attempts with a subsequent successful 4624 event.

Scenario 2: Privilege Escalation

- Monitor Event ID 4720 for new accounts.
- Monitor Event ID 4732 for changes to privileged local groups.
- Verify whether the changes were authorised.

Scenario 3: Persistence Detection

- Monitor Event ID 7045 for newly installed services.
- Investigate unknown service names and executable paths.
- Verify whether the service installation was legitimate.

Scenario 4: Anti-Forensic Activity

- Monitor Event ID 1102.
- Identify which account cleared the Security log.
- Investigate preceding suspicious activities and other available logs.

8. Learning Outcomes

After completing this practical, I learned:

- How Windows records successful and unsuccessful logon attempts.
- How to identify authentication activities through Event IDs 4624 and 4625.
- How to create and monitor local user accounts.
- How to detect changes in the local Administrators group.
- How Windows records newly installed services.
- How Security log clearing is recorded.
- How to filter Windows Event Logs using Event IDs.
- How Windows event monitoring supports SOC investigations.
- How suspicious activities can be identified through event correlation.

9. Conclusion

In this practical, I explored Windows Event Logging by generating six important Windows events and analysing their corresponding records using Event Viewer.

I investigated failed and successful logons, user account creation, administrator group membership changes, service installation, and Security log clearing.

This exercise helped me understand how Windows records security-related activities and how SOC analysts can use these records to detect suspicious behaviour, investigate privilege escalation, identify persistence mechanisms, and recognise attempts to remove investigation evidence.

These concepts will be useful in future practical activities involving endpoint monitoring, Windows event analysis, and SIEM/IDS-based threat detection.